

Reducing Heat Dissipation Requirements by Content Brightness Modulation for Lighter Headset

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ABSTRACT

Weight reduction in a headset is critical for improving XR usability. This poster investigates whether components related to thermal management (e.g., heat sinks) can be reduced or eliminated, in a PCVR (or thin client type) headset through display brightness and thus temperature modulation. We share findings from several tests, measuring temperature ranges (and differences) among various operating conditions, including the existence of the heat sink, especially on the display. Results have shown a potential for reduction of approximately 2–3% of weight for the lightest PCVR headset (about 125 g).

Index Terms: Virtual Reality, Head-mounted Display, Thin headset, OLED Display, Thermal management

1 INTRODUCTION

Thin and lightweight headsets aim to improve comfort by reducing the physical size and weight of standalone headsets, putting emphasis on the usability, while relying on wireless remote computation for performance [1]. Off-loading rendering and computation can reduce on-device power consumption and heat dissipation, potentially relaxing the need for thermal-management hardware such as cooling fans or heat sinks.

Unlike the standalone headsets, which are already significantly heavy (>500 g) [2], for the thin headset, even a relatively small amount of weight reduction can be significant. Thus, this poster investigates whether thermal-management components (e.g., heat sinks) can be reduced or eliminated through display brightness (i.e., content brightness, pixel intensity) and temperature modulation. We present findings from several tests, measuring temperature ranges (and differences) among various conditions – by display component type, display brightness levels, and the existence of the heat sink.

2 EXPERIMENT

Measurement setup. We measured headset temperatures (LCD or OLED) using a hand-held thermal camera (FLIR ONE Pro) in a tabletop (not-worn) setup. Camera distance and viewing angle were fixed at 50 cm using physical markers to reduce sensitivity to small pose changes (see Figure 1(a)). For each condition, we captured 5 shots at the fixed measurement points (see Figure 1(b,c)) and report the median of the 5-shot peaks (robust to outliers).

Test Conditions. There were two factors: (1) headset (or display component) type and (2) content brightness levels. The former factor had two levels – (a) fast-switching LCD (e.g., Meta Quest 2 [2]) and (b) OLED (e.g., Bigscreen Beyond [3]), and the latter, 5 levels – (a) black image (0 brightness), (b) white image (full brightness), (c) typical regular content (mid/high range brightness), (d) Level

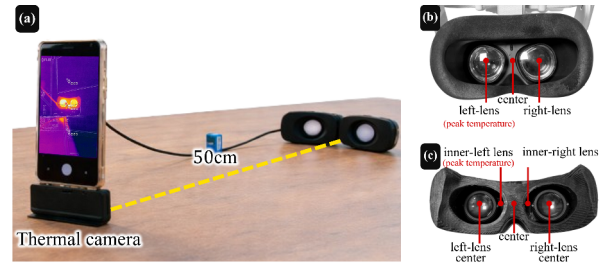


Figure 1: Measurement setup and measurement regions. (a) Smartphone based tabletop thermal-imaging setup with the camera positioned 50 cm from the headset. (b) Display-side measurement points for the LCD-based headset (three points). (c) Display-side measurement points for the OLED-based headset (five points).

1 blurred content (brightness dispersed, $\text{boxblur}=10:1$, radius 10 pixels, 1 iteration) and (e) Level 2 blurred content (brightness further dispersed, $\text{boxblur}=20:3$, radius 20 pixels, 3 iterations).

In addition, the LCD-based headset was tested with the cooling fan (its removal resulted in a system halt), while the OLED-based headset, with both the heat sink attached and removed. In particular, OLED displays can plausibly produce different thermal loads depending on content brightness and emitting area, and we anticipated that certain content (e.g., low-brightness scenes) could yield lower display-side temperatures. On the other hand, the LCD requires constant backlight, expectedly limiting any temperature difference on the display by content brightness modulation.

Procedure and controls.

For each condition, the stimulus was displayed continuously for 60 minutes, followed by thermal image capture. To prevent thermal carry-over (residual heat), we ensured at least 1 hour of cool-down between conditions. Measurements were also initiated at least 1 hour after charging to exclude charging-related heating effects.

Measurement regions.

We measured consistent regions on the display side. For the LCD-based headset, we recorded temperatures at three points (see Figure 1(b)). For the OLED-based headset, we recorded temperatures at five points (see Figure 1(c)). Note that our measurements focused on display-side components (rather than the overall device or processing hardware) to explore the possibility of reducing or removing heat-related components in the display assembly.

3 RESULTS

3.1 LCD-based Headset case

As expected, the LCD-based headset showed only small display-side temperature differences (most probably due to the constant backlight) across content conditions (approximately 3–4 °C for Black vs. White (e.g., Black: 32 °C vs. White: 35 °C)). The video clarity (three levels of blurring) did not produce a clear separation either. In other words, the dispersed brightness had no effect.

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3.2 OLED-based Headset case

Content brightness/dispersion effect: The OLED-based headset showed a clear dependence on content brightness under *both* heat-sink configurations. For Black vs. White images, the peak display-side temperature differed by approximately 11–13 °C under both heat-sink configurations. In contrast, for the video clarity conditions, peak temperatures were similar across the three clarity (blur) levels (see Figure 2). This indicates that the brightness dispersion had no effect, as the overall brightness did not change.

Heat-sink attachment effect: Across conditions, peak display-side temperatures were similar between the heat-sink-attached and heat-sink-removed configurations, with differences typically within 1–3 °C (see Figures 2 and 3). Because this range is comparable to thermal-imaging variability from small environmental or geometric changes, we treat it as within measurement uncertainty (i.e., no reliably detectable heat-sink benefit) in our tabletop 60-minute protocol. This is consistent with the possibility that the heat sink mainly serves as a guard against rare extreme conditions; thus, for typical usage with content brightness modulation, removing the heat sink (about 3 g for the tested headset) may be feasible.

4 DISCUSSION AND IMPLICATIONS

Taken together, the results align with the expected behavior of the two display technologies: a backlit LCD system tends to exhibit relatively stable thermal output across different displayed content, whereas a self-emissive OLED system can exhibit strongly content-dependent heating. In our OLED tests, content brightness (and emitting area) was a much stronger driver of display-side temperature than blur level, as evidenced by the large Black–White contrast (11–13 °C) and the negligible differences across video clarity (or brightness dispersion) levels.

These findings suggest that content-aware modulation may be a practical complementary lever for managing display-side temperature in OLED-based headset designs. For example, this could be achieved by reducing average brightness or by restricting the effective field of view (FOV) via peripheral black masking to reduce the effective emissive area. Prior work on subtle dynamic FOV modification reports reduced VR sickness while maintaining presence, suggesting that such masking can remain perceptually acceptable when carefully tuned [4]. However, the thermal effect of applying explicit brightness/FOV modulation policies has not yet been directly quantified in our setup, and warrants validation.

Although preliminary yet, these results motivate software-side approaches that may relax thermal-hardware requirements and enable lighter, slimmer packaging in follow-up designs.

5 CONCLUSION

In the OLED-based headset, Black vs. White images produced a large peak-temperature contrast (approximately 11–13 °C), well above the thermal camera’s practical uncertainty, under *both* heat-sink configurations. This large contrast supports the use of brightness-aware and emissive-area-aware modulation (e.g., peripheral black masking to restrict the effective FOV) to reduce display-side heat dissipation requirements. Within our tested regime (tabletop setup and 60-minute exposure), removing the heat sink changed peak temperature by only approximately 1–3 °C; we did not observe a clearly detectable thermal benefit from the heat sink in this protocol.

Taken together, these results suggest that, for typical usage with such modulation, the tested headset can potentially operate without the heat sink, enabling lighter and slimmer packaging (about 3 g reduction for this device). Although 3 g is modest, it is an initial step that may enable more perceptible cumulative reductions in follow-up designs (e.g., downsizing cooling hardware and slimmer packaging); prior work suggests that even small weight changes can affect comfort in relatively light integrated HMDs [5].

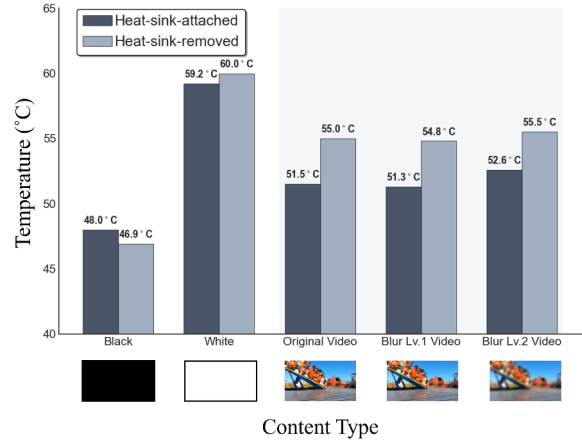


Figure 2: Peak display-side temperatures of the OLED-based headset across five content conditions with/without the heat sink. Black vs. White produces a large peak-temperature contrast (11–13 °C) under both configurations. In contrast, three video clarity (brightness dispersion) conditions show no meaningful separation.

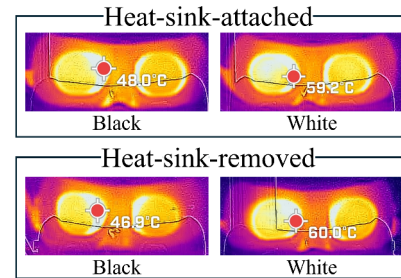


Figure 3: Peak temperatures for OLED-based headset with/without the heat sink and for 0 (black) and full (white) brightness images.

To generalize beyond our not-worn tabletop tests, future work should map the no-heat-sink operating envelope by testing longer-duration *worn* use, higher ambient temperature, more extreme content, and stronger masking to assess whether peak temperatures can be maintained in the high-40 °C range.

ACKNOWLEDGMENTS

This work was partly supported by the NRF grant (RS-2025-00514411) and also IITP-ITRC grant (IITP-2022-RS-2022-00156354), all funded by the MSIT of Korea.

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